

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

***Artificial Intelligence
and Society***



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 3: Unsupervised & Reinforcement Learning

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Review Lecture 2

- *Question: How can we measure how well a supervised learning algorithm performs?*

Answer: For classification, we use accuracy (proportion of correct predictions) or the F_1 score (harmonic mean of positive predictive value and sensitivity). For regression, we use the root mean squared error.

- *Question: What is Bias (underfitting) and Variance (overfitting)?*

Answer: Bias: a systematic deviation of model predictions from the true value.

Variance: a spread of predictions for similar input values.

- *Question: What data types exist and what problems can occur with data?*

Answer: Data types: discrete and continuous numbers, classes, images, text, time series. Problems: missing data, outliers, "forbidden" values.

Overview of Today's Lecture

[content for further reading]

- 3.1 Introduction to Unsupervised Learning
- 3.2 Distance Measures
- 3.3 Dimensionality Reduction
- 3.4 Algorithms for Unsupervised Learning
- 3.6 Reinforcement Learning

K. Backhaus et al., [Multivariate Analysemethoden](#),
Chapter 8, Pages 435-497 (German).

Computer vision for dummies, [The Curse of Dimensionality in classification](#).

C. M. Bishop, [Pattern Recognition and Machine Learning](#),
Chapter 9, Pages 423-430.

[A step-by-step look at Alpha Zero and Monte Carlo Tree Search](#)

Learning Objectives

- Know application areas of unsupervised learning and understand algorithms for unsupervised learning.
- Understand how to measure the distance between data points and reduce dimensionality.
- Know the functional principle and application areas of reinforcement learning.